

MODULE 06

Backup & Restore

Protect NovaMart's data with full, differential, and log backups — and practice restoring before disaster strikes.

Recovery Models

Backup Types

Restore Strategy

Day 2 · Morning · 2 hours
Module 06: Backup & Restore

1 / 8

Slide 1 of 8

Module 06: Backup & Restore

Module	Day	Duration	Lab
06	2 — Morning	2 hours	Lab 06

A backup is your last line of defence against data loss. Accidental table drops, hardware failure, ransomware, and human error are all resolved the same way: restore from a clean backup. This module teaches you how to choose the right recovery model, build a layered backup strategy, and practice restoring so that when disaster strikes you can act quickly and confidently.

NovaMart runs a production database. By the end of this module you will have performed a full backup, a differential backup, and a transaction log backup — then simulated a disaster and restored the complete database.



Test Your Restores — Not Just Your Backups

An untested backup is not a backup — it is a hypothesis. Run a restore drill at least once per quarter on a test instance. Confirm that the database comes online, data is intact, and the

restore completes in the expected time.

What This Module Covers

- **Recovery Models** Simple, Full, and Bulk-Logged — how each affects what you can restore.
- **Backup Types** Full, differential, and transaction log backups — when and why to use each.
- **T-SQL Backup** BACKUP DATABASE and BACKUP LOG commands with key options.
- **Restore Sequence** NORECOVERY vs RECOVERY, STOPAT for point-in-time, and REPLACE.
- **Lab 06** Full disaster simulation: backup → drop table → restore → verify.

What You Will Learn

By the end of Module 06 you will be able to:

-  Explain the difference between Simple and Full recovery models
-  Perform a full database backup using SSMS and T-SQL
-  Perform a differential backup to capture incremental changes
-  Back up and restore the transaction log to a point in time
-  Restore the NovaMart database from a backup file and verify data

Module 06: Backup & Restore

2 / 8

Slide 2 of 8

Learning Objectives

By the end of Module 06 you will be able to:

- **Explain the difference between Simple and Full recovery models** — and set the recovery model with ALTER DATABASE.
- **Perform a full database backup using SSMS and T-SQL** — BACKUP DATABASE ... TO DISK with FORMAT, INIT, COMPRESSION.
- **Perform a differential backup to capture incremental changes** — BACKUP DATABASE ... WITH DIFFERENTIAL.
- **Back up and restore the transaction log to a point in time** — BACKUP LOG; RESTORE LOG ... WITH STOPAT.
- **Restore the NovaMart database from a backup chain and verify data** — applying full → differential → log WITH NORECOVERY / RECOVERY in sequence.

Recovery Models

The recovery model controls how much of the transaction log is retained — and what you can restore.

SIMPLE

Log truncated automatically at each checkpoint.
No log backups possible.
Maximum data loss = changes since last full or diff backup.
Good for: development, test, non-critical databases.

FULL

Full transaction log kept until a log backup is taken.
Point-in-time restore is possible.
Requires regular log backups to prevent log file growth.
Required for: NovaMart production database.

BULK-LOGGED

Minimally logs bulk operations (BULK INSERT, SELECT INTO).
Reduces log growth during large data loads.
Point-in-time restore not available during bulk period.
Typically used as a temporary switch during large imports.

```
ALTER DATABASE NovaMart SET RECOVERY FULL;
```

NovaMart should always use FULL recovery.

Recovery Models

The recovery model is a database-level setting that controls how much transaction log history SQL Server retains, which in turn determines what you can restore. Choose the wrong model and you may lose hours of data even when backups ran successfully.

Model	Log Truncation	Log Backups	Point-in-Time Restore	Best For
SIMPLE	Automatic (at checkpoint)	Not possible	No	Dev/test, read-only databases, low-criticality
FULL	Only via log backup	Required	Yes	Production databases where zero data loss matters
BULK-LOGGED	Only via log backup	Required	Limited	Temporary switch during large bulk operations to reduce log growth



NovaMart Must Use FULL Recovery

NovaMart is a production database. Always set it to FULL recovery. SIMPLE recovery truncates the

log automatically, making transaction log backups impossible and point-in-time restore unavailable.

T-SQL

```
-- Check the current recovery model
SELECT name, recovery_model_desc
FROM sys.databases
WHERE name = 'NovaMart';

-- Set to FULL recovery
ALTER DATABASE NovaMart SET RECOVERY FULL;

-- Verify the change
SELECT name, recovery_model_desc
FROM sys.databases
WHERE name = 'NovaMart';
```

Log File Growth Warning

Under FULL recovery the transaction log grows continuously until a log backup is taken. If you set FULL recovery but never schedule log backups, the log file will grow until the drive is full. Always pair FULL recovery with a log backup schedule.

Backup Types

Use a combination of backup types to balance storage cost against recovery time and data loss.

FULL Backup

Backs up the entire database — data pages + enough log to make it consistent.
Self-contained: can restore from this one file alone.
Typically run: weekly or nightly (small DBs).

DIFFERENTIAL Backup

Backs up all pages changed since the last FULL backup.
Grows larger until the next full backup resets the baseline.
Typically run: nightly or every few hours.

TRANSACTION LOG Backup

Backs up the active portion of the transaction log.
Enables point-in-time restore (STOPAT).
Truncates the log — required in FULL recovery to control log growth.
Typically run: every 15–60 min in production.

Strategy for NovaMart: Full (Sunday) → Differential (Mon–Sat) → Log every 30 min

Module 06: Backup & Restore 4 / 8

Slide 4 of 8

Backup Types

SQL Server provides three complementary backup types. Use them in combination to balance storage cost, backup window, and recovery time objective (RTO).

Type	What It Captures	Self-Contained	Typical Schedule
Full Backup	Entire database — all data pages plus enough log to ensure consistency	Yes — can restore from this file alone	Weekly (large DBs) or nightly (small DBs)
Differential Backup	All pages changed since the last full backup (grows until next full)	No — requires the last full backup	Nightly or every few hours
Transaction Log Backup	Active log since the last log backup; truncates log after backup	No — requires full + prior log chain	Every 15–60 minutes in production

NovaMart Recommended Strategy

Full backup every Sunday → Differential backup Monday through Saturday → Transaction log backup every 30 minutes around the clock.

- **Full (Sun) Baseline** — enables all restores for the week. Keep for 4 weeks.

- **Diff (Mon–Sat)** Reduces restore time vs log-only chain. Keep for 1 week.
- **Log (every 30m)** Point-in-time precision. Keep for 48 hours minimum.



First Log Backup After Setting FULL Recovery

After switching from SIMPLE to FULL recovery, take a full backup immediately. Until the first full backup is taken after the model change, SQL Server keeps the database in "pseudo-SIMPLE" mode and log backups are not possible.

Performing Backups via T-SQL

T-SQL backup commands are scriptable, schedulable, and the foundation for SQL Agent jobs.

```
-- Full backup
BACKUP DATABASE NovaMart
TO DISK = 'C:\SQLBackup\NovaMart_Full.bak'
WITH FORMAT, INIT,
     NAME = 'NovaMart Full Backup',
     COMPRESSION, STATS = 10;

-- Differential backup
BACKUP DATABASE NovaMart
TO DISK = 'C:\SQLBackup\NovaMart_Diff.bak'
WITH DIFFERENTIAL, INIT,
     NAME = 'NovaMart Differential Backup',
     COMPRESSION, STATS = 10;

-- Transaction log backup
BACKUP LOG NovaMart
TO DISK = 'C:\SQLBackup\NovaMart_Log.bak'
WITH INIT, NAME = 'NovaMart Log Backup', STATS = 10;
```

FORMAT, INIT

Overwrite existing backup file. Remove INIT to append.

COMPRESSION

Reduces backup file size — recommended for speed and storage.

STATS = 10

Print progress every 10% — useful for large databases.

WITH DIFFERENTIAL

Captures only pages changed since the last full backup.

BACKUP LOG

Only valid when recovery model is FULL or BULK-LOGGED.

Backup to network

Replace disk path with \\server\share\file.bak for UNC paths.

VERIFY after backup

RESTORE VERIFYONLY FROM DISK = '...' — check integrity.

Module 06: Backup & Restore

5 / 8

Slide 5 of 8

Performing Backups via T-SQL

T-SQL backup commands are the foundation for all automation. They are used directly in ad-hoc scripts, SQL Agent job steps, and maintenance plan tasks.

T-SQL

```
-- — Full backup —————
BACKUP DATABASE NovaMart
TO DISK = 'C:\SQLBackup\NovaMart_Full.bak'
WITH FORMAT,          -- overwrite file; create new backup set
     INIT,             -- overwrite existing backup sets in the file
     NAME = 'NovaMart Full Backup',
     COMPRESSION,      -- smaller file, faster I/O
     STATS = 10;       -- print progress every 10 %

-- — Differential backup —————
BACKUP DATABASE NovaMart
TO DISK = 'C:\SQLBackup\NovaMart_Diff.bak'
WITH DIFFERENTIAL,
     INIT,
     NAME = 'NovaMart Differential Backup',
     COMPRESSION, STATS = 10;

-- — Transaction log backup —————
```



```

BACKUP LOG NovaMart
TO DISK = 'C:\SQLBackup\NovaMart_Log.bak'
WITH INIT,
    NAME = 'NovaMart Log Backup',
    STATS = 10;

```

Key Backup Options Reference

Option	Effect	When to Use
FORMAT	Creates a new media set — overwrites all existing backup sets on the file	First backup to a new file or when recycling a file
INIT	Overwrites existing backup sets in the media set (without new media header)	Nightly overwrites of yesterday's backup file
NOINIT	Appends to the existing backup sets in the file	Multiple backups in one file (audit trail)
COMPRESSION	Compresses the backup — reduces size 50–80 %	Always on unless restoring to pre-SQL 2008 R2
STATS = 10	Prints completion percentage every 10 % in the Messages pane	Long-running backups; script visibility
COPY_ONLY	Ad-hoc backup that does not affect the differential baseline	Manual backup before a risky change without disrupting schedule
VERIFY ONLY	Checks backup integrity without restoring	RESTORE VERIFYONLY FROM DISK = '...' after every backup

Restoring a Database

A backup is worthless unless you have practised the restore. Always test your backups.

```
-- Step 1: Restore the full backup (WITH NORECOVERY keeps DB in
-- restoring state)
RESTORE DATABASE NovaMart
FROM DISK = 'C:\SQLBackup\NovaMart_Full.bak'
WITH NORECOVERY, STATS = 10;

-- Step 2: Apply differential backup
RESTORE DATABASE NovaMart
FROM DISK = 'C:\SQLBackup\NovaMart_Diff.bak'
WITH NORECOVERY, STATS = 10;

-- Step 3: Apply log backup - bring online
RESTORE LOG NovaMart
FROM DISK = 'C:\SQLBackup\NovaMart_Log.bak'
WITH RECOVERY;

-- Optional: restore to a point in time
-- WITH RECOVERY, STOPAT = '2026-05-08T09:30:00';
```

- NORECOVERY**
Leaves DB in restoring state — required when more backups follow.
- RECOVERY**
Brings database online — use only on the FINAL restore step.
- STOPAT**
Point-in-time recovery — only available with FULL recovery model.
- REPLACE option**
Add WITH REPLACE to overwrite an existing database by same name.
- MOVE option**
WITH MOVE 'NovaMart' TO '...mdf', MOVE 'NovaMart_log' TO '...ldf'
- Tail-log backup**
Before restoring a live DB, back up the tail of the active log first.
- Verify before restore**
RESTORE HEADERONLY FROM DISK='...' to see backup sets on file.

Module 06: Backup & Restore

6 / 8

Slide 6 of 8

Restoring a Database

Restoring requires applying backups in the correct order: full first, then differential (if available), then log backups in sequence. Every step except the last uses WITH NORECOVERY to keep the database in a "Restoring" state that accepts further backups. The final step uses WITH RECOVERY to bring the database online.

T-SQL

```
-- — Step 1: Restore the full backup —————
RESTORE DATABASE NovaMart
FROM DISK = 'C:\SQLBackup\NovaMart_Full.bak'
WITH NORECOVERY,      -- keep DB in Restoring; more backups follow
      STATS = 10;

-- — Step 2: Apply differential backup —————
RESTORE DATABASE NovaMart
FROM DISK = 'C:\SQLBackup\NovaMart_Diff.bak'
WITH NORECOVERY,
      STATS = 10;

-- — Step 3: Apply log backup - bring database online —————
RESTORE LOG NovaMart
FROM DISK = 'C:\SQLBackup\NovaMart_Log.bak'
WITH RECOVERY;
```

```
-- -- Optional: point-in-time recovery _____
-- Replace WITH RECOVERY above with:
-- WITH RECOVERY, STOPAT = '2026-05-15T09:30:00';
```

Restore Options Reference

Option	Effect
NORECOVERY	Database stays in Restoring state — required when more backups will be applied
RECOVERY	Brings database online — use only on the FINAL restore step
STOPAT	Point-in-time restore — requires FULL recovery model and a log backup that spans the target time
REPLACE	Overwrites an existing database of the same name — use with caution
MOVE ... TO	Relocates data and log files to different paths on the target server



Take a Tail-Log Backup Before Restoring a Live Database

If you need to restore a live database without losing the most recent transactions, first run:
BACKUP LOG NovaMart TO DISK = '...' WITH
NO_TRUNCATE;

This captures the active log even if the database is damaged. Then apply it as the final step of the restore chain.

LAB 06

NovaMart Disaster Recovery Simulation

Duration: ~60 min · Tool: SSMS · Database: NovaMart

STEPS

- 1 Confirm NovaMart is in FULL recovery: `SELECT name, recovery_model_desc FROM sys.databases`
- 2 Create backup folder: `C:\SQLBackup` (or verify it exists)
- 3 Run full backup T-SQL to NovaMart_Full.bak — note completion time
- 4 Insert a test row: `INSERT INTO Employee VALUES (...)` — note the time
- 5 Run differential backup to NovaMart_Diff.bak
- 6 Insert another test row — note time
- 7 Run log backup to NovaMart_Log.bak
- 8 Simulate disaster: `DROP TABLE Employee` — confirm table gone
- 9 Restore full → differential → log WITH RECOVERY
- 10 Verify Employee table and all rows are back

VALIDATION

- ✓ Backup files created in `C:\SQLBackup`
- ✓ DROP TABLE removes Employee
- ✓ Restore completes without error
- ✓ Employee table exists after restore
- ✓ All rows present including inserted rows

Module 06: Backup & Restore 7 / 8

Slide 7 of 8

Lab 06: NovaMart Disaster Recovery Simulation

Goal	Duration	Tool	Database
Full backup chain, disaster simulation, and complete restore	~60 min	SSMS	NovaMart

Prerequisites

- NovaMart database in FULL recovery model.
- `C:\SQLBackup\` folder created on the server (or create it in step 2).
- Employee table exists with at least 3 rows (from Lab 04).

Part A: Set Up and Perform Backups

1. Confirm FULL recovery model:

`SELECT name, recovery_model_desc FROM sys.databases WHERE name = 'NovaMart';`

Expected: `recovery_model_desc` shows FULL. If SIMPLE, run: `ALTER DATABASE NovaMart SET RECOVERY FULL;`

2. Create the backup folder if it does not exist.

In PowerShell: `New-Item -ItemType Directory -Path C:\SQLBackup -Force`

Expected: C:\SQLBackup\ folder exists.

3. Run a full backup:

```
BACKUP DATABASE NovaMart
TO DISK = 'C:\SQLBackup\NovaMart_Full.bak'
WITH FORMAT, INIT, NAME = 'NovaMart Full', COMPRESSION, STATS = 10;
```

Expected: Messages pane shows 100% and "BACKUP DATABASE successfully processed N pages."

4. Note the current time. Insert a test row:

```
INSERT INTO dbo.Employee (FirstName, LastName, DeptID)
VALUES ('Disaster', 'TestRow', 1);
SELECT * FROM dbo.Employee;
```

Expected: New row visible. Note the time after the insert.

5. Run a differential backup:

```
BACKUP DATABASE NovaMart
TO DISK = 'C:\SQLBackup\NovaMart_Diff.bak'
WITH DIFFERENTIAL, INIT, NAME = 'NovaMart Diff', COMPRESSION, STATS = 10;
```

Expected: Differential backup completes successfully.

6. Insert a second test row:

```
INSERT INTO dbo.Employee (FirstName, LastName, DeptID)
VALUES ('Second', 'TestRow', 2);
```

Expected: Row inserted.

7. Run a log backup:

```
BACKUP LOG NovaMart
TO DISK = 'C:\SQLBackup\NovaMart_Log.bak'
WITH INIT, NAME = 'NovaMart Log', STATS = 10;
```

Expected: Log backup completes. "Second TestRow" is now in the log backup.

Part B: Simulate Disaster

8. Simulate accidental data loss:

```
DROP TABLE dbo.Employee;
SELECT * FROM dbo.Employee; -- should fail
```

Expected: Error: "Invalid object name 'dbo.Employee'." Table is gone.

Part C: Restore

9. Restore the full backup (NORECOVERY — more backups follow):

```
RESTORE DATABASE NovaMart
FROM DISK = 'C:\SQLBackup\NovaMart_Full.bak'
WITH NORECOVERY, STATS = 10;
```

Expected: Database enters Restoring state. It is not yet accessible.

10. Apply the differential backup:

```
RESTORE DATABASE NovaMart
FROM DISK = 'C:\SQLBackup\NovaMart_Diff.bak'
WITH NORECOVERY, STATS = 10;
```

Expected: Differential applied successfully.

11. Apply the log backup and bring the database online:

```
RESTORE LOG NovaMart
FROM DISK = 'C:\SQLBackup\NovaMart_Log.bak'
WITH RECOVERY;
```

Expected: Database comes online. "RESTORE DATABASE successfully processed."

12. Verify full recovery:

```
SELECT * FROM dbo.Employee;
```

Expected: Employee table exists with all original rows AND both TestRow rows inserted in steps 4 and 6.

Validation Checklist

- ☐ NovaMart_Full.bak, NovaMart_Diff.bak, NovaMart_Log.bak all exist in C:\SQLBackup\
- ☐ DROP TABLE confirmed the table was gone before restore
- ☐ Restore chain applied in correct order without errors
- ☐ Employee table accessible after restore with RECOVERY
- ☐ Both TestRow rows present — confirming differential and log were applied

Troubleshooting

Symptom	Likely Cause	Resolution
BACKUP LOG fails	Recovery model is SIMPLE	ALTER DATABASE NovaMart SET RECOVERY FULL; then take a full backup first
RESTORE fails: "exclusive"	SSMS is connected to	Set context to master before

access"	NovaMart	restoring: USE master; then RESTORE
Restore: "too new to be restored"	Log backup is older than diff backup	Apply backups in strict sequence: full → diff → log
Database stays in Restoring state	RECOVERY not applied on final step	Run: RESTORE DATABASE NovaMart WITH RECOVERY;

Day 2 Progress

Module 06 complete — NovaMart data is now protected.

M06 ✓ Backup & Restore Full, Diff, Log backups · Recovery models · Disaster recovery sim	M07 Database Maintenance Index rebuild/reorganize · Statistics · DBCC CHECKDB
M08 SQL Server Agent Jobs, schedules, alerts · Automate nightly backup	M09 Monitoring & Perf. Activity Monitor · DMVs · Blocking & deadlocks
M10 Troubleshooting Connection errors · Error logs · Performance bottlenecks	M11 Import / Export Wizard · CSV/Excel · BCP · ETL concepts

Module 06: Backup & Restore
Next up: Module 07 — Database Maintenance
8 / 8

Slide 8 of 8

Day 2 Progress — Module 06 Complete

NovaMart data is now protected. You have built a complete backup strategy and proven it works with a full disaster simulation. The remaining Day 2 modules build on this foundation — automation, monitoring, and troubleshooting.

Module	Topic	Status
06	Backup & Restore	✓ Complete — full backup chain, disaster simulation, restore verified
07	Database Maintenance	Next — index rebuild/reorganize, statistics, DBCC CHECKDB
08	SQL Server Agent	Upcoming — automate the nightly backup job created today
09	Monitoring & Performance	Upcoming — Activity Monitor, DMVs, blocking and deadlocks
10	Basic Troubleshooting	Upcoming — connection errors, error log, performance first response
11	Data Import & Export	Upcoming — Wizard, BULK INSERT, BCP, ETL concepts

i**What Comes Next: Automate This in Module 08**

In Module 08 (SQL Server Agent) you will turn today's manual backup commands into a scheduled SQL Agent job that runs every night at 02:00. Module 07 first covers database maintenance — keeping indexes and statistics healthy so queries remain fast after the data loads you performed in Module 04.